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Further analysis of the expectations hypothesis using very short-term rates

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Abstract

Longstaff [Longstaff, F., 2000. The term structure of very short-term rates: new evidence for the expectations hypothesis. Journal of Financial Economics 58, 397–415] finds support for the expectations hypothesis at the very short end of the repurchase agreement (repo) term structure while other studies find calendar-time-based regularities cause rejection of the expectations hypothesis. Using Longstaff's methods on a sample of repo rates that pre-dates Longstaff's sample, we reject the expectations hypothesis for every maturity. The pre-Longstaff-sample repo data comes from a time period where the behavior of short-term interest rates is similar to the long-run average behavior of short-term interest rates. Our results imply that expectations hold when rates are less volatile and/or that we may be entering a period of lower volatility.

JEL classification: E43; G21

Keywords: Expectations hypothesis; Term structure; Repurchase agreements

1. Introduction

Longstaff (2000) uses the term structure of very short-term (overnight through 3-month) general collateral government repurchase agreement (repo) rates to test empirically the expectations hypothesis and reports evidence across all terms that strongly supports this hypothesis.[1] Downing and Oliner (2007) extend Longstaff's analysis by examining the very short end of the term structure of interest rates in the commercial paper market and reject the expectations hypothesis. However, after controlling for year-end rate increases in commercial paper rates, they find that commercial paper rates generally conform to the expectations hypothesis.[2]

Longstaff (2000) uses the overnight repo rate as the short rate and the term repo rates as the long rates in his various estimations. Griffiths and Winters (1997b) examine term (1-week through 3-month) general collateral government repo rates and find a year-end rate increase for 1-week, 2-week, 3-week, and 1-month repos. They suggest that the identified rate pattern is most consistent with a year-end preferred habitat for liquidity. Given the identified year-end rate increase in term repo rates and the

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[1] Longstaff (2000) defines the expectation hypothesis as $E[R_{t+n}|\Omega_t] = Y_t(n) + a_n$ where R_{t+n} is the average short rate from time t to time $t+n$, Ω_t is the information set, Y_t (n) is the long rate, and a_n is a constant term premium. Longstaff notes that the results from his unconditional tests are consistent with $a_n = 0$ and thus, consistent with the pure form of the expectations hypothesis. However, he does not specifically constrain his methods to test only for pure expectations.

[2] See Musto (1997) and Griffiths and Winters (2005) for discussions of the year-end rate increase in commercial paper rates.

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Downing and Oliner (2007) results, we revisit Longstaff (2000) to determine if his results are general or sample specific.

Using repo rates for most of the Longstaff sample period (May 1991 to July 1997) acquired from a different source than Longstaff (2000), we find mixed evidence for the expectations hypothesis. Specifically, we find three of six alpha estimates are significant, two of six beta estimates are significant, and we reject the joint hypothesis of all coefficients equal to zero for all six maturities both within equations and jointly for all equations. We then extend our analysis to a pre-sample set of repo rates (February 1984 to May 1991) used by Griffiths and Winters (1997a,b). The multivariate results from the pre-sample data reject the expectations hypothesis across all terms, which suggest that Longstaff's results do not generalize to the Griffiths and Winters sample period. A lack of generalization is consistent with the findings of Lange et al. (2003) and Downing and Oliner (2007). When calendar based variables are added, the December indicator is mostly significant for the in-sample estimation, and 1- and 2-week estimations continue to show rejection of the expectations hypothesis. When using a representative ARCH-in-mean model, we find that the expectations hypothesis is rejected for every maturity for pre-, in-sample, and a combined sample with an indicator variable.

When we combine the pre- and in-sample periods, the in-sample indicator variable reveals significantly higher alpha estimates and lower beta estimates across every maturity. We continue to reject the expectations hypothesis and we show that volatility is lower during the in-sample period than during the period studied by Griffiths and Winters (1997b). We conclude that Longstaff's strong support for the expectations hypothesis does not generalize well.

2. Data and term premiums

Bloomberg does not have archived data prior to 1991 so, in order to test the generality of Longstaff's results, we employ data obtained from the Chicago Mercantile Exchange. These data (1-week, 2-week, 3-week, and one, two, and three- month maturities as well as overnight repo rates) are used because they allow for testing in an alternative sample period as well as overlapping most of Longstaff's sample period. These data do not correspond exactly with Longstaff's Bloomberg source, but the lowest correlation is 0.99095 for each maturity.[3]

Longstaff (2000) uses repo rates from May 21, 1991 to October 15, 1999. Griffiths and Winters (1997b) use repo rates from February 2, 1984 to December 31, 1991 and find strong evidence of year-end rate increases in 1-week through 1-month general collateral government repos. In addition, Griffiths and Winters (1997a) use repo rates from February 2, 1984 to January 31, 1995 and find a Federal Reserve bank settlement effect in overnight general collateral government repo rates.[4] Because the majority of the sample period covered in both Griffiths and Winters studies predates the Longstaff sample period, the data used by Griffiths and Winters provide an appropriate sample for alternative sample tests on repo data. We extend the Griffiths and Winters data to July of 1997 using additional data acquired from the Chicago Mercantile Exchange (CME). The data end in 1997 because the CME stopped collecting daily repo data at that time. We divide the sample at May 20, 1991 and refer to the sample period on or before May 20, 1991 as our pre-sample period and then apply Longstaff's primary methods to the data for our alternative-sample tests. Tests on data that overlap Longstaff's sample period are referred to as in-sample tests.

2.1. Term premiums

Using the Chicago Mercantile repo data, we calculate the in-sample term premiums for the overlapping period, which we report in Panel B of Table 1. We provide the Longstaff (2000) term premiums in Panel A for comparison. The in-sample period has similar average rates, but the term premiums are much higher for our data. In fact, our data indicate positive and significant term premiums for every maturity except 1-week instruments using Hansen and Hodrick (1980) adjusted standard errors.

The summary statistics reported in Panel C show that the rates and term premiums in the pre- sample data are quite different from the original sample period both from Longstaff (2000) and our data source. First, the pre-sample test period average term repo rates are generally between 7.85% and 7.99% while the rates for the Longstaff sample data generally range from 4.70% and 4.75% and in our in-sample data range from 4.60% to 4.65%. Second, the pre-sample term premiums begin at an insignificant 0.02 basis points (bps) for 1-week repos and increase monotonically to 12.68 bps for 3-month repos, while in the Longstaff data the term premiums start at 0.56 bps for 1-week repos and increase monotonically to 3.19 bps for 3-month repos. It is noteworthy that the Hansen–Hodrick adjusted significance levels are typically just below the 10-percent level even though the pre-sample term premiums are over three times larger than the Longstaff sample period. The insignificant term premiums are consistent with pure expectations on average. However, the substantial differences in the mean term premiums suggest that repo rates across the two periods are quite different and that univariate results

[3] Specifically, the correlations are overnight equal to 0.99095, 1-week equal to 0.99607, 2-week equal to 0.99738, and 1-month equal to 0.99776. Three-week, 2-month, and 3-month repo rates are not available from our Bloomberg source, thus we cannot calculate correlations for these terms.

[4] Griffiths and Winters (1997a,b) both commence on February 2, 1984 as this was the beginning of a bi-weekly settlement period instituted by the Federal Reserve and government repos are allowed as substitutes for fed funds. Prior to this date, settlement had been on a weekly basis. For consistency with the earlier studies, we also commence our pre-sample period on February 2, 1984.

Table 1
Average repurchase agreement (repo) terms premiums

Repo term	Average term repo rate	Average overnight repo rate	Term premium (%)	*t*-statistic	*N*
Panel A: Longstaff (2000) results (from Table 2, p. 405)					
One-week	4.7062	4.7006	0.0056	1.35	2091
Two-week	4.7118	4.7028	0.0090	1.48	2086
Three-week	4.7152	4.7038	0.0114	1.45	2081
One-month	4.7248	4.7045	0.0203	1.91	2074
Two-months	4.7313	4.7065	0.0248	1.41	2052
Three-months	4.7417	4.7098	0.0319	1.15	2031
Panel B: In-sample data from CME data source					
One-week	4.6037	4.5983	0.0054	1.18	2214
Two-week	4.6130	4.5952	0.0178***	2.57	2207
Three-week	4.6198	4.5920	0.0279***	3.04	2200
One-month	4.6249	4.5890	0.0359***	3.19	2184
Two-months	4.6403	4.5726	0.0677***	3.90	2161
Three-months	4.6504	4.5573	0.0931***	3.57	2131
Panel C: Pre-sample data					
One-week	7.8564	7.8562	0.0002	0.02	2633
Two-week	7.8846	7.8595	0.0251	1.50	2613
Three-week	7.8972	7.8570	0.0402*	1.87	2623
One-month	7.9013	7.8566	0.0447	1.53	2616
Two-months	7.9454	7.8642	0.0812	1.50	2586
Three-months	7.9854	7.8585	0.1268*	1.70	2553

The repo data in Panel A are from May 1991 to October 1999 (Longstaff, 2000 data), in Panel B from May 1991 to July 1997, and the data in Panel C are from February 1984 to May 1991 for the pre-sample period. The *t*-statistics are based on the Hansen and Hodrick (1980) covariance estimate where the lag equals the overlap in observations as in Longstaff (2000).
*,**,*** Term premium is significantly different from 0 at the 10-, 5-, and 1-percent level, respectively.

could be unable to discover the statistical differences in the data.

2.2. Possible sample differences

The in-sample term premiums and significance are much different in our data source as compared to Longstaff (2000). There are two possible causes for this difference. First, the overlap of our data and the Longstaff sample period is not complete with our data having over 2 years fewer observations than the Longstaff's sample. A second possible distinction is the way that we handle non-trading days for our repo rates (Longstaff does not report how he handles non-trading days). In our case, we follow industry convention and count any trading day preceding a non-trading period as an n-period rate that spans the trading horizon. That is, if a rate is a Friday "overnight" rate, then the repo actually pays interest for 3 days, and we duplicate the rate over the weekend. We also span non-trading holidays in this same manner, and thus our sample size averages about 30 days per month per 1-week rates while the Longstaff (2000) sample averages about 18 observations per month for 1-week rates. Since weekends are the most common non-trading period and Friday rates are lower on average (see Griffiths and Winters, 1997a) excluding these weekend days would tend to bias the term premium downward. The different observations per month account for the larger *N* with our in-sample data, even though our sample period covers about 2 years less of calendar time.

The summary statistics are markedly different for the in-sample period versus the pre-sample period and we do not know which set of results is more representative of historically normal market conditions. Using weekly T-bill yields from January 1960 to January 2003, we find that the sample period covered by the Griffiths and Winters (1997b) sample has characteristics similar to the averages from 1960 to 2003.[5] The Longstaff (2000) sample period corresponds to a period of low volatility and hence, greater predictability. However, using data from November 1999 to January 2003, which postdates the Longstaff sample period, we find that the most recent data also have lower volatility than average, suggesting that Longstaff's sample could be consistent with a new market equilibrium.

3. Tests using Longstaff's (2000) basic regression analysis

In this section, we use standard methods similar to Longstaff (2000) to investigate the (pure) expectations hypothesis for short-term repo rates. Downing and Oliner (2007) initially reject the expectations hypothesis using short-term commercial paper rates. However, after controlling for year-end rate increases in commercial paper rates, they find that commercial paper rates generally conform to the expectations hypothesis. Griffiths and Winters (1997b, 2005) find that year-end rates increase in commercial paper

[5] Details available upon request.

and repos and suggest that the rate increases in both instruments are consistent with a preferred habitat for liquidity at year-end. Developed by Modigliani and Sutch (1966), the preferred habitat hypothesis explains twists or humps in the yield curve, which implies a change in the slope of the yield curve. Longstaff's model does not include controls for possible year-end rate increases and thus does not allow for changes in the slope of the yield curve. Accordingly, we re-visit Longstaff's analysis to determine its generality.

3.1. Longstaff's primary methods with our extensions

Longstaff's primary analysis involves the estimation of the following equation:

$$R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \varepsilon_{t+n}, \tag{1}$$

where R_{t+n} is the average of the short rate over n days and $Y_t(n)$ is the long rate of term n days. In Eq. (1) $\alpha = 0$ and $\beta = 0$ is consistent with the pure expectations hypothesis, and more particularly if the joint hypothesis holds for both equaling zero. However, if a non-zero term premium is present, $\beta = 0$ is consistent with the expectation hypothesis. Longstaff focuses on the more general expectations hypothesis, but identifies results that support pure expectations. We follow Longstaff's procedures initially and then add calendar dummy variables as well as use a representative time-series model shown to be well-fitted to interest rate data.

We estimate Eq. (1) using ordinary least squares (OLS) with both Hansen and Hodrick (1980) and Newey and West (1987) standard errors. The Newey–West standard errors are superior to the OLS test statistic as they correct for serially correlated error terms and heteroskedasticty, but we report Hansen–Hodrick test statistics to be consistent with Longstaff as well as using the most conservative test statistics. We also estimate sup-bound confidence intervals similar to Cavanagh et al. (1995) and Downing and Oliner (2007) because our regression is likely to have nearly integrated regressors due to the nature of the overlapping observations. The sup-bound adjustment provides for very conservative local-to-unity critical values to account for the persistence and finite sample bias inherent in using data with such substantial overlap (see Downing and Oliner, 2007 for an application to testing the short-term expectations hypothesis). Hence, our results are biased towards not rejecting the (pure) expectations hypothesis.

3.2. Estimation of Eq. (1) on the in-sample data

The primary test of the expectations hypothesis used by Longstaff (2000) is an OLS estimation of Eq. (1) with Hansen and Hodrick (1980) test statistics. Recall, that the null hypothesis for the pure expectations hypothesis is $\alpha = 0$ and $\beta = 0$, the null hypothesis for the expectations hypothesis with a term premium is $\beta = 0$ and that Longstaff's estimates fail to reject both null hypotheses for each term from 1-week to 3-months.

Table 2, Panel A contains our results from the same basic model as Longstaff (2000) but uses our data source over a slightly shorter sample period than Longstaff. We also present the sup-bound lower and upper confidence intervals that account for possible persistence and finite sample biases. Estimates indicate rejection of the expectations hypothesis for 1-week and 2-week rates, but provide support for the other maturities. The intercept is significantly positive for 1-week repo rates as well. We conjecture that the difference between our results and Longstaff (2000) are due to either the shorter time period than Longstaff, or to the difference in accounting for non-trading days as discussed previously.

Table 2 also contains the more general Newey and West (1987) t-statistics which correct for heteroskedasticity in addition to our specification of lagging the covariances the same number of lags as the repo term. These t-statistics are uniformly higher, so we focus on the more conservative Hansen and Hodrick (1980) procedure for the remainder of the paper.

In Panel B of Table 2 we present the estimates for the in-sample data with the addition of calendar indicator variables. We add these variables because Downing and Oliner (2007) find that commercial paper rates generally conform to the expectations hypothesis only after controlling for year-end rate increases in commercial paper rates. Similarly, Griffiths and Winters (1997a,b) find calendar effects in repo rates. Thus, we modify Eq. (1) to include the identified calendar-based effects in overnight and short-term repo rates and estimate the following modified equation on our in-sample repo rates:

$$\begin{aligned} R_{t+n} - Y_t(n) = {} & \alpha_n + \beta_n Y_t(n) + \phi \text{SETTLE} \\ & + \rho \text{DECEMBER} + \omega \text{HOLIDAY} \\ & + \mu \text{MONTHEND} + \varepsilon_t, \end{aligned} \tag{2}$$

with calendar dummies for settlement Wednesday (SETTLE), maturity past the December month-end (DECEMBER), the day before a holiday (HOLIDAY), and the last trading day of the month-end (MONTHEND).

Results show the same pattern for the intercept and slope estimates with 1-week and 2-week rates having significant coefficients for both, and the intercept remaining significant for 3-week rates. The most consistent result for the calendar variables is that December is positive and significant in every regression except for 3-month rates. The increase in December repo rates is consistent with Griffiths and Winters (1997b).

The last column of Table 2 in both Panels A and B contains the chi-squared test statistic for the joint hypothesis that all coefficients are equal to zero within the equation, consistent with the pure expectations hypothesis. Results indicate that the joint hypothesis is rejected for every maturity at the five-percent level of significance or better, for the model *without* calendar indicators. Similarly, the joint hypothesis that all coefficients are equal to zero, including the calendar indicators in Panel B is also rejected. Note

Table 2
In-sample ordinary least squares estimation

Term	α Estimate	Hansen–Hodrick *t*-statistics	Newey–West *t*-statistics	β Estimate	Hansen–Hodrick *t*-statistics	Newey–West *t*-statistics	Lower 95% critical value	Upper 95% critical value	Joint test
Panel A: In-sample results from May 1991 to July 1997									
One-week	0.0530***	2.69	3.12	−0.0103**	−2.38	−2.75	−1.65	2.03	8.6**
Two-week	0.0689**	2.48	3.00	−0.0111**	−1.82	−2.22	−1.70	2.05	12.3***
Three-week	0.0751**	1.98	2.43	−0.0148	−1.26	−1.55	−1.69	2.11	12.1***
One-month	0.0581	1.24	1.47	−0.0048	−0.46	−0.55	−1.72	2.05	12.1***
Two-months	0.0559	0.83	0.89	0.0025	0.16	0.18	−1.73	2.10	17.5***
Three-months	0.0244	0.29	0.32	0.0148	0.70	0.83	−1.64	2.18	15.4***

	α Estimate	β Estimate	Settle	December	Pre-holiday	Month-end	Joint test
Panel B: In-sample results from May 1991 to July 1997 with calendar-based variables and Hansen–Hodrick t-statistics							
One-week	0.0546***	−0.0103**	−0.0177**	0.0200**	−0.0384	−0.0121	22.7***
	(2.79)	(−2.35)	(−2.51)	(2.00)	(−0.86)	(−1.39)	
		[−1.66, 2.01]					
Two-week	0.0678**	−0.0111**	−0.0067	0.0201**	0.0199	0.0139	15.7**
	(2.49)	(−1.85)	(−1.44)	(2.10)	(1.07)	(1.36)	
		[−1.67, 2.13]					
Three-week	0.0752**	−0.0102	−0.0094**	0.0294**	0.0032	−0.0079	14.3**
	(1.99)	(−1.27)	(−2.03)	(2.26)	(0.16)	(−0.87)	
		[−1.69, 2.13]					
One-month	0.0600	−0.0048	−0.0040	0.0388***	−0.0363	0.0070	19.2***
	(1.25)	(−0.45)	(−0.80)	(2.96)	(−1.17)	(0.00)	
		[−1.70, 2.07]					
Two-months	0.0587	0.0025	−0.0151*	0.0338**	−0.0348	0.0109	44.8***
	(0.85)	(0.16)	(−1.79)	(2.02)	(−0.71)	(0.00)	
		[−1.70, 2.12]					
Three-months	0.0285	0.0148	−0.0082	−0.0066	−0.0370	−0.0091**	43.9***
	(0.32)	(0.70)	(−1.06)	(−0.55)	(−0.65)	(−1.80)	
		[−1.64, 2.17]					

	OLS coefficients	OLS coefficients with calendar dummy regressions
Panel C: Joint Wald tests using simultaneous equations on sample data from May 1991 to July 1997		
Pure expectations (all coefficient = 0)	46.6***	25.8***
General expectations (slope and calendar coefficient = 0)	18.8***	27.0***

Note: The OLS model is $R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \phi\text{SETTLE} + \rho\text{DECEMBER} + \omega\text{HOLIDAY} + \mu\text{MONTHEND} + \varepsilon_t$ with R_{t+n} equals to the average overnight rate for n days and $Y_t(n)$ equal to the term repo rate of maturity n at time t. The null hypothesis is $\alpha = 0$ and $\beta = 0$. SETTLE equals one on settlement Wednesday and zero otherwise; DECEMBER equals one if the repo matures across the end of the year and zero otherwise; HOLIDAY equals one on the day before a market closed holiday and zero otherwise; MONTHEND is one at month end and zero otherwise.

Standard errors are either Hansen and Hodrick (1980) or Newey and West (1987) with lags equal to the term of the repo. The upper and lower critical values and the bracketed term beneath the beta *t*-statistic are confidence intervals for a persistent regressor as in Cavanagh et al. (1995). The joint test in Panel C is a joint test that all coefficients are jointly equal to zero.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

that the rejection of the joint hypothesis is in the context of most of the individual coefficients being insignificant, which supports rejection of the pure expectations model for this sample.

Panel C of Table 2 presents the Wald statistics to test the joint hypothesis that all coefficients are equal to zero for all equations simultaneously. The statistics indicate that pure expectations, with all coefficients equal to zero, and general expectations with only the slope coefficients equal to zero, are soundly rejected.[6]

3.3. Estimation of Eq. (1) on the pre-sample data

Table 3 presents the pre-sample estimates for the basic model (Eq. (1)), the model with calendar indicators (Eq. (2)), and the joint hypothesis tests.

3.3.1. Results from Eq. (1) without controls for seasonal rate changes

The estimates in Panel A of Table 3 indicate that α is negative and significant at the 5% level or better for each term from 1-week to 3-months and that β is positive and significant at the 10% level or better for each term from 1-week to 3-months. Therefore, the pre-sample results reject the expectations hypothesis for each term. It should also be noted that rejection of the expectations hypothesis holds using Hansen–Hodrick t-statistics evaluated against the sup-bound conditions for each term. That is, when we use the most conservative test statistic and account for persistence and finite sample bias, the expectations hypothesis is clearly rejected.

3.3.2. Results from Eq. (1) with controls for seasonal rate changes

The results in Panel A of Table 3 exclude the calendar variables found important in prior research, which could bias the results. Panel B of Table 3 contains the OLS regression results including the calendar-based effects. The focus in this panel is twofold: first, whether or not the alpha and beta estimates are still significant and second, whether any of the calendar-effects are significant. As shown, the alpha and beta estimates are little changed in this panel as compared to Panel A. Every alpha estimate is negative and significant and declines monotonically through the maturity of the term repos. The beta estimates are also consistent and increase monotonically throughout. We conclude that the rejection of the expectations hypothesis is not due to omitting the calendar-based variables found important in the Griffiths and Winters papers. While only a few of the calendar-based parameter estimates are significant, we continue to reject the expectations hypothesis at all terms. One important exception is that for 1-month, 2-month, and 3-month estimates, the t-statistics are not outside the 95% confidence intervals for the sup-bound conditions, although they are outside the 90% interval (not shown).[7]

In the last column of Panels A and B of Table 3, we present the chi-squared statistics for the joint hypothesis of all coefficients equal to zero. In all cases for the model without calendar dummies, the joint hypothesis is rejected at the 10% level or better. In the case for the model containing calendar indicators, the 1-month and 3-month rates do not reject the pure expectations hypothesis. Not rejecting the null hypothesis for the calendar model individual coefficients in Panel B coupled with the results for the joint hypothesis within equations suggests that both the expectations hypothesis and pure expectations hypothesis are holding for one- and 3-month rates. However, the results in Panel C for the joint tests for all equations simultaneously indicate rejection of both the pure and general expectations hypotheses at greater than the 1-percent level of significance. Failure to reject the expectations hypothesis for 1-month and 3-month rates is in contrast to the results in Panel A without calendar variables and indicates the importance of including these variables when studying the short end of the term structure.

3.4. ARCH-in-mean additional tests

Interest rates have been shown to have auto-regressive conditional heteroskedasticity (ARCH) in prior research.[8] Many forms of this model have economic foundations that drive the model choice but, in our case, our focus is not on a model specification search. Rather, we choose a model that has performed well in controlling for ARCH effects in prior term structure research. The model we choose is from Engle et al. (1987) who use an ARCH-in-mean model to test for the expectations hypothesis. The model is:

[6] We thank an anonymous referee for suggesting this technique which accounts for the fact that we are conducting multiple tests and therefore using all relevant parameters in a six-equation system.

[7] In an earlier version of this paper, we estimated a vector autoregression GARCH model and simulated shocks to the system using the variance for the model as a GARCH process with factor innovations as in Bekaert et al. (1997) and in Longstaff's (2000). Unfortunately, as shown in these papers, there is a potential bias due to small sample size. The model includes over 100 coefficients to describe the dynamics of the short rate and there is no consideration of alternative specifications. The adjusted p-values that this procedure generates are only valid if the underlying short rate model is an accurate description of reality. The results are available upon request and are markedly different from the results obtained by Longstaff (2000) but consistent with other overnight rate variance estimates as in Cyree and Winters (2001) and Cyree et al. (2003). The smallest ARCH term for our sample is greater than 0.49, while the largest in Longstaff's results is 0.27. Further, the increased persistence in variance indicates that the sample periods are much different. We thank an anonymous referee for pointing out several difficulties with this method.

[8] See Christiansen (2005) for a recent survey of (G)ARCH models as applied to interest rate research.

Table 3
Pre-Sample ordinary least squares estimation

Term	α Estimate	Hansen–Hodrick t-statistics	Newey–West t-statistics	β Estimate	Hansen–Hodrick t-statistics	Newey–West t-statistics	Lower 95% critical value	Upper 95% critical value	Joint test
Panel A: Pre-sample data from February 1984 to May 1991									
One-week	−0.2219***	−3.42	−4.22	0.0283***	3.32	4.10	−1.645	2.332	12.0***
Two-week	−0.3208***	−3.11	−3.84	0.0439***	3.20	3.79	−1.645	2.237	10.4***
Three-week	−0.3482**	−2.52	−3.12	0.0492**	2.67	3.33	−1.676	2.177	8.0**
One-month	−0.3970**	−2.13	−2.67	0.0559**	2.25	2.83	−1.645	2.244	5.6*
Two-months	−0.6256**	−1.97	−2.33	0.0890*	2.10	2.50	−1.645	2.475	5.0*
Three-months	−0.8646**	−2.04	−2.28	0.1242*	2.17	2.45	−1.645	2.399	5.4*

	α Estimate	β Estimate	Settle	December	Pre-holiday	Month-end	Joint test
Panel B: Pre-sample results from February 1984 to May 1991 with calendar-based variables and Hansen–Hodrick t-statistics							
One-week	−0.2191***	0.0281***	−0.0061	−0.0966	0.0068	0.0013	12.8**
	(−3.39)	(3.31)	(−0.58)	(−1.15)	(0.41)	(0.08)	
		[−1.64, 2.32]					
Two-week	−0.3243***	0.0438***	−0.0035*	0.0263	0.0214	0.0777***	41.9***
	(−3.15)	(3.20)	(−0.32)	(0.28)	(1.21)	(3.88)	
		[−1.69, 2.27]					
Three-week	−0.3474**	0.0491**	−0.0033	−0.0215	0.0352*	0.0175	11.6*
	(−2.51)	(2.67)	(−0.43)	(−0.29)	(1.66)	(1.15)	
		[−1.68, 2.18]					
One-month	−0.3917**	0.0557*	−0.0100	−0.0350	0.0175	−0.0124	7.5
	(−2.09)	(2.24)	(−0.95)	(−0.41)	(1.09)	(−0.01)	
		[−1.64, 2.24]					
Two-months	−0.6338**	0.0886*	−0.0001	0.1224	0.0240	0.0197	47.0***
	(−2.02)	(2.17)	(−0.02)	(0.80)	(1.11)	(1.42)	
		[−1.64, 2.48]					
Three-months	−0.8679**	0.1233*	−0.0016	0.1149	0.0055	0.0159	7.1
	(−2.09)	(2.23)	(−0.19)	(0.75)	(0.27)	(0.79)	
		[−1.64, 2.41]					

	OLS coefficients	OLS coefficients with calendar dummy regressions
Panel C: Joint Wald tests using simultaneous equations on pre-sample data from February 1984 to May 1991		
Pure expectations (all coefficient = 0)	27.4***	78.6***
General expectations (slope and calendar coefficient = 0)	28.4***	47.1***

Note: The OLS model is $R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \phi\text{SETTLE} + \rho\text{DECEMBER} + \omega\text{HOLIDAY} + \mu\text{MONTHEND} + \varepsilon_t$ with R_{t+n} equals to the average overnight rate for n days and $Y_t(n)$ equal to the term repo rate of maturity n at time t. The null hypothesis is $\alpha = 0$ and $\beta = 0$. SETTLE equals one on settlement Wednesday and zero otherwise; DECEMBER equals one if the repo matures across the end of the year and zero otherwise; HOLIDAY equals one on the day before a market closed holiday and zero otherwise; MONTHEND is one at month end and zero otherwise. Standard errors are either Hansen and Hodrick (1980) or Newey and West (1987) with lags equal to the term of the repo. The upper and lower critical values and the brackets beneath the beta t-statistic are confidence intervals for a persistent regressor as in Cavanagh et al. (1995). The joint test in Panel C is a joint test that all coefficients are jointly equal to zero.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

$$R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \phi \text{SETTLE} + \rho \text{DECEMBER} + \omega \text{HOLIDAY} + \mu \text{MONTHEND} + \delta \log h_t + \varepsilon_t, \quad (3)$$

where the variance is specified as:

$$h_t = a_0 + a_1 \sum_{\tau=1,4} w_\tau \varepsilon_{t-1}^2 + c\text{SETTLE} + d\text{DECEMBER} + f\text{HOLIDAY} + g\text{MONTHEND} \quad (4)$$

with $w_\tau = (6 - \tau)/15$ to weight the past innovations across the days of the week. Note that the variance in the mean term is contemporaneous and in logs as in Engle et al. (1987). Engle et al. find that this model works best in a class of models tested while investigating the expectations hypothesis.[9] The indicator variables are the same as in Eq. (2) however we add the log of the variance into the mean in Eq. (3).

The ARCH-in-mean specification allows for a variance in the mean equation and controls for the effect of errors if, the average overnight rates are far from the expected value. This model also has the additional feature of a smooth transition for lagged errors in the variance equation and the log of variance in the mean.

3.4.1. In-sample ARCH-in-mean results

We continue to use the extended Chicago Mercantile data for our in-sample period.[10] The Chicago Mercantile data are only available to July 1997 and thus does not cover the entire Longstaff (2000) sample period to October 1999. We use the ARCH-in-mean model of Engle et al. (1987) on the Chicago Mercantile data during Longstaff's sample period with calendar dummies for settlement Wednesday (SETTLE), maturity past the end of December (DECEMBER), the day before a holiday (HOLIDAY), and the last trading day of the month-end (MONTHEND).

Table 4 contains the results of the ARCH-in-mean estimation and reveals a significant intercept and slope coefficient in all terms except 2 weeks. The monotonic relation over time has disappeared, likely due to the influence of the variance in the mean equation which has significant coefficients for 2-week, 1-month, and 2-month repos. The calendar indicators are mostly inconsistent with the exception that the December indicators are negative when significant, but are not significant in every case. The within equation joint hypothesis that all mean coefficients are jointly equal to zero is soundly rejected as shown by the chi-squared statistics in the bottom row of Panel A. The joint test is also highly significant in Panel B where the equations are estimated simultaneously. These results indicate that the expectations hypothesis is not holding for data that covers most of the in-sample period when accounting for ARCH and variance-in-mean effects.

3.4.2. Pre-sample ARCH-in-mean results

Table 5 contains the estimates of the ARCH-in-mean model for the pre-sample period. The alpha estimates are negative and significant for every term except 1-week. The beta coefficient is positive and highly significant for all terms to maturity. The volatility-in-mean term is also significant for 1-week, 3-week, and 1-month instruments and indicates that volatility increases the term premium in addition to the importance of accounting for the errors in this specification. Here, the monotonic relation across the terms to maturity holds for both the intercept and slope. The within equation chi-squared statistics (Panel A) reject the joint hypothesis that all coefficients are equal to zero as do the simultaneous joint test in Panel B. After controlling the volatility-in-mean effect, we continue to reject the expectations hypothesis in the pre-sample data.

The variance effects in this model are mixed with some models exhibiting long persistence in shocks to the variance as evidenced by ARCH terms greater than one. Most of the indicator variables are inconsistent, although in general, settlement Wednesdays have higher variance and month-ends tend to have lower variance when there are significant estimates.

3.4.3. Entire sample results

The evidence so far indicates that Longstaff's sample period may have been a period of more stable rates and low volatility and therefore caused acceptance of the (pure) expectations theory because of an unusually quiet sample period. Figs. 1 and 2 plot the 1-week and 1-month repos relative to the overnight repo rate respectively, with these spreads averaged across each month. The plots clearly show that the volatility was less during the sample period beginning in 1991.

To investigate whether or not the sample period is different statistically, we use the ARCH-in-mean model with the addition of indicator variables for the in-sample period. The indicator represents a possible regime shift in the intercept and slope in the mean equation and for a shift in the volatility for the variance equation.

The repurchase agreement data from the Chicago Mercantile cover the period from February 1984 to July 1997, which contains most of the Longstaff (2000) sample period as well as all of our pre-sample period. The model is:

$$R_{t+n} - Y_t(n) = \alpha_n + \alpha'_n \text{DUMMY} + \beta_n Y_t(n) + \beta'_n Y_t(n)\text{DUMMY} + \phi \text{SETTLE} + \rho \text{DECEMBER} + \omega \text{HOLIDAY} + \mu \text{MONTHEND} + \delta \log h_t + \varepsilon_t. \quad (5)$$

[9] While there are many variants of (G)ARCH models that could be evaluated using these data, this type of investigation is beyond the scope of this paper. Rather, our aim is to discover if our rejection of the expectations hypothesis is due to well known time-series effects. Thus, we use a model that has been shown to be well specified.

[10] Recall that the CME data is the only data available in both the pre-sample and in-sample period as discussed in the subsequent section.

Table 4
ARCH-in-mean estimation with Hansen and Hodrick (1980) standard errors for the in-sample period

Variable	One-week		Two-week		Three-week		One-month		Two-month		Three-month	
	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics
Panel A: In-sample period from May 1991 to July 1997												
α	0.0509***	3.10	0.0028	0.13	0.1004***	5.21	0.1517***	8.38	−0.6399***	−40.54	0.1437***	4.78
β	−0.0127***	−6.47	−0.0145***	−5.92	−0.0130***	−5.94	−0.0165***	−4.77	0.0835***	34.36	−0.0101**	−2.24
SETTLE	−0.0081**	−2.23	−0.0054	−0.95	−0.0083*	−1.94	0.0286***	5.74	−0.0210***	−3.00	−0.0059	−0.85
DECEMBER	−0.0641***	−11.41	0.0069	0.53	−0.0397***	−8.81	−0.0866***	−8.33	−0.1322***	−10.11	−0.0015	−0.15
HOLIDAY	0.0003	0.06	0.0340***	2.79	0.0147**	2.39	0.0054	0.47	0.1398***	9.48	−0.0110	−1.42
MONTHEND	0.0025	0.39	0.0887***	8.15	−0.0039	−0.67	0.0127	1.47	0.0176	1.61	−0.0252**	−2.41
δ	−0.0030	−1.29	−0.0154***	−4.61	0.0044	1.26	0.0067***	4.87	0.0083***	2.49	0.0055	1.07
a_0	0.0011***	10.69	0.0008***	10.59	0.0007***	7.48	0.0009***	7.19	0.0013***	4.73	0.0010***	8.12
a_1	0.9915***	19.67	0.9224***	20.19	1.0782***	40.22	1.1014***	10.75	1.1014***	13.74	0.9758***	13.10
SETTLE	−0.0001	−0.26	0.0000	−0.07	0.0003	0.85	0.0005	1.23	−0.0001	−0.86	0.0006***	2.75
DECEMBER	0.0034**	2.07	0.0008*	1.66	0.0014***	2.97	−0.0005***	−2.08	−0.0012***	−4.29	−0.0010***	−6.24
HOLIDAY	−0.0014***	−6.19	0.0005	0.47	−0.0011**	−2.51	−0.0005	−1.05	0.0206*	1.73	−0.0001	−0.10
MONTHEND	0.0013	1.17	0.0084***	5.73	0.0000	0.02	0.0031***	2.78	−0.0012***	−3.31	0.0012	1.14
Joint test	271.667***		379.740***		330.651***		2342.746***		5156.528***		163.194***	
Panel B: Joint Wald tests using multivariate ARCH-M equations on in-sample data from May 1991 to July 1997												
Pure expectations (all coefficient = 0)								5103.1***				
General expectations (slope and calendar coefficient = 0)								757.4***				

Note: The model is: $R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \phi$ SETTLE + ρDECEMBER + ωHOLIDAY + μMONTHEND + $\delta \log h_t + \varepsilon_t$ with R_{t+n} equals to the average overnight rate for n days and $Y_t(n)$ equal to the term repo rate of maturity n at time t. The null hypothesis is $\alpha = 0$ and $\beta = 0$.

The variance is specified as: $h_t = a_0 + a_1 \sum_{\tau=1,4} w_\tau \varepsilon_{t-1}^2 + c$SETTLE + dDECEMBER + fHOLIDAY + gMONTHEND with $w_\tau = (6-\tau)/15$ to weight the past innovations across the days of the week as in Engle et al. (1987). SETTLE equals one on settlement Wednesday and zero otherwise; DECEMBER equals one if the repo matures across the end of the year and zero otherwise; HOLIDAY equals one on the day before a market closed holiday and zero otherwise; MONTHEND is one at month end and zero otherwise.

Standard errors are Hansen and Hodrick (1980) robust standard errors with the lagged error equal to the term of the repurchase agreement for each maturity.

*,**,*** Significant at the 10%, 5%, and 1% level, respectively.

Table 5
ARCH-in-mean estimation with Hansen and Hodrick (1980) standard errors for the pre-sample period

Variable	One-week		Two-week		Three-week		One-month		Two-month		Three-month	
	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics	Estimate	t-statistics
Panel A: Pre-sample period from February 1984 to May 1991												
α	−0.0505	−1.48	-0.1465^{***}	−4.48	-0.1551^{**}	−2.54	-0.2707^{***}	−11.43	-0.5801^{***}	−39.06	-0.6737^{***}	−11.58
β	0.0080^{**}	2.60	0.0162^{***}	5.19	0.0295^{***}	6.06	0.0373^{***}	17.39	0.0751^{***}	24.27	0.0980^{***}	26.60
SETTLE	−0.0002	−0.03	−0.0030	−0.46	0.0027	0.33	0.0167^{**}	1.99	0.0008	0.08	-0.0123^{**}	−1.98
DECEMBER	-0.1058^{***}	−3.76	-0.1047^{***}	−11.87	-0.0527^{**}	−2.09	-0.0926^{***}	−10.84	-0.1487^{***}	−6.28	0.0504^{*}	1.85
HOLIDAY	0.0155	1.53	-0.0278^{***}	−2.66	0.0023	0.21	−0.0082	−0.60	0.0364^{***}	3.07	0.0013	0.09
MONTHEND	0.0168^{*}	1.86	0.0335^{***}	3.27	0.0177	0.81	−0.0181	−1.23	0.0137	1.15	0.0525^{***}	3.50
δ	0.0080^{*}	1.84	−0.0037	−1.03	0.0212^{**}	2.03	0.0056	1.03	0.0048	1.12	0.0394^{***}	3.88
a_0	0.0033^{***}	14.03	0.0027^{***}	9.35	0.0044^{***}	16.12	0.0019^{***}	7.94	0.0024^{***}	9.56	0.0024^{***}	4.81
a_1	0.9737^{***}	21.46	1.0891^{***}	21.74	0.9462^{***}	16.44	1.0595^{***}	13.45	1.0686^{***}	17.03	1.0282^{***}	10.22
SETTLE	0.0047^{***}	3.54	−0.0009	−1.36	−0.0012	−1.36	0.0023^{**}	2.59	-0.0022^{***}	−3.85	0.0008	0.95
DECEMBER	0.0352	1.59	0.0008	0.43	0.0034^{**}	2.38	0.0016^{*}	1.67	−0.0006	−0.56	-0.0018^{***}	−2.68
HOLIDAY	-0.0014^{**}	−1.99	-0.0028^{***}	−16.52	−0.0012	−0.64	0.0010	0.47	−0.0004	−0.37	−0.0003	−0.38
MONTHEND	-0.0025^{***}	−3.55	0.0001	0.08	0.0029^{*}	1.77	0.0032^{*}	1.69	-0.0024^{***}	−6.25	-0.0030^{***}	−7.11
Joint test	26.956^{***}		329.6207^{***}		54.486^{***}		585.726^{***}		4779.583^{***}		3354.285^{***}	
Panel B: Joint Wald tests using multivariate ARCH-M equations on pre-sample data from February 1984 to May 1991												
Pure expectations (all coefficient = 0)								$322{,}534^{***}$				
General expectations (slope and calendar coefficient = 0)								$280{,}877^{***}$				

Note: The model is: $R_{t+n} - Y_t(n) = \alpha_n + \beta_n Y_t(n) + \phi$ SETTLE $+ \rho$DECEMBER $+ \omega$HOLIDAY $+ \mu$MONTHEND $+ \delta \log h_t + \varepsilon_t$ with R_{t+n} equals to the average overnight rate for n days and $Y_t(n)$ equal to the term repo rate of maturity n at time t. The null hypothesis is $\alpha = 0$ and $\beta = 0$.

The variance is specified as: $h_t = a_0 + a_1 \sum_{\tau=1,4} w_\tau \varepsilon_{t-1}^2 + c$SETTLE $+ d$DECEMBER $+ f$HOLIDAY $+ g$MONTHEND with $w_\tau = (6 - \tau)/15$ to weight the past innovations across the days of the week as in Engle et al. (1987). SETTLE equals one on settlement Wednesday and zero otherwise; DECEMBER equals one if the repo matures across the end of the year and zero otherwise; HOLIDAY equals one on the day before a market closed holiday and zero otherwise; MONTHEND is one at month end and zero otherwise. Standard errors are Hansen and Hodrick (1980) robust standard errors with the lagged error equal to the term of the repurchase agreement for each maturity.

*,**,*** Significant at the 10%, 5%, and 1% level, respectively.

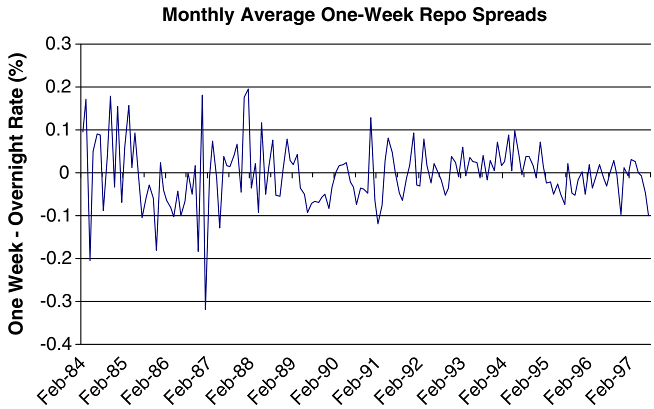

Fig. 1. Monthly average 1-week repo spreads.

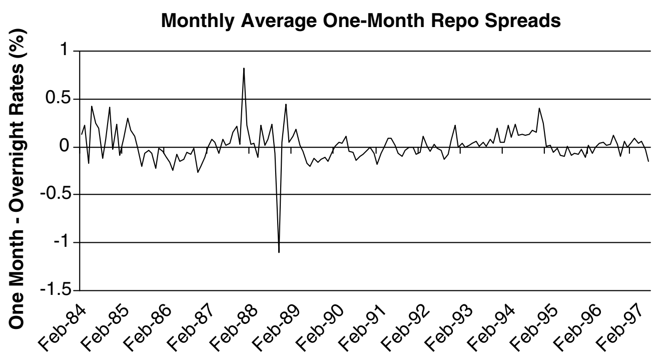

Fig. 2. Monthly average one-monthly repo spreads.

The variance is specified as:

$$h_t = a_0 + a_0'\text{DUMMY} + a_1 \sum_{\tau=1,4} w_\tau \varepsilon_{t-1}^2 + c\text{SETTLE} + d\text{DECEMBER} + f\text{HOLIDAY} + g\text{MONTHEND}. \quad (6)$$

where DUMMY is equal to one after December 1991 and zero otherwise, and all other variables are as defined in Eqs. (2)–(4).

In effect, we use this model to test if the sample period that coincides with most of Longstaff's is significantly different. Table 6 presents the results of the ARCH-in-mean estimates for the full sample. Two important results emerge. First, the regime shift dummy variable is positive and highly significant for the intercept in every case, and the slope interaction term is negative and highly significant for all terms to maturity. The second important result is that all of the alpha estimates are significant, and all slope estimates are positive and significant. Likewise, the shift term in the variance indicates lower volatility for the Longstaff (2000) in-sample period in accordance with Figs. 1 and 2. Joint tests, both within each equation and simultaneously, reject both the pure and general expectations hypothesis at greater than the 1-percent level. Together, these results imply that the expectations hypothesis is not supported, and that the sample period that contains most of the Longstaff (2000) sample period is clearly different.

4. Discussion and additional analysis

The results from the pre-sample data strongly reject the expectations hypothesis for all terms (1-week through 3-months) after adjusting for possible calendar and time-series variance effects. These results are contrary to Longstaff's (2000) results, which support the expectations hypothesis for each term. An additional test suggests that the Longstaff sample period is significantly different from the pre-sample period. Therefore, the question that remains is which set of results best represents the general market condition?

Our first choice is to perform our analysis on the term structure of another US money market with rates available in terms from overnight to 3 months. The only other US money market with readily available overnight rates is the fed funds market. However, the overnight fed funds rate exhibits strong bank settlement Wednesday rate change and volatility effects, so the overnight fed funds rate would not be appropriate for testing the expectations hypothesis. An additional problem with fed funds is the limited availability of term rates and limited market entry.

Our second choice is to collect data from a long time period to examine whether the time period covered by the pre-sample data or the time period covered by the Longstaff (2000) data is more representative. The money market with the most readily available data over a long time period with different maturities is the T-bill market. We collect 3-month and 6-month T-bill yields weekly from January 1960 through January 2003 and calculate the weekly spread between the bills. To get a feel for the general nature of the term structure, we calculate the average spread and the standard deviation of the spread for each decade. We exclude from our calculations data from the highly volatile period when the Federal Reserve experimented with targeting M1 (October 1979 through October 1982). These summary statistics are reported in Table 7.

We find that the decade with the highest spread and most volatility is the 1970s. The decade that is closest to the overall average spread and standard deviation is the decade of the 1980s (after excluding the period when the Fed targeted M1). This time period is similar to the time period covered by our pre-sample data. From the decades in which we have 10 years of data, the 1990s has the smallest spread and the second lowest standard deviation. These results provide two insights about our results on the expectations hypothesis using the repo term structure. First, the pre-sample data period corresponds to the decade that is closest to average for the period from January 1960 to January 2003. Accordingly, one can conclude that rejection of the expectations hypothesis using the pre-sample data is the more representative result. Second, the Longstaff (2000) sample period corresponds to the decade with the lowest spread and below average volatility and hence, greater predictability. Low volatility is a sample characteristic that would contribute to accepting the expectations hypothesis. Finally, the limited data from the beginning of the 2000s show a continuation of the low volatility during the

Table 6
ARCH-in-mean estimation with regime shift for the entire sample period and Hansen and Hodrick (1980) standard errors

Variable	One-week		Two-week		Three-week		One-month		Two-month		Three-month	
	Estimate	*t*-statistics	Estimate	*t*-statistics	Estimate	*t*-statistics	Estimate	*t*-statistics	Estimate	*t*-statistics	Estimate	*t*-statistics
Panel A: Sample period from February 1984 to July 1997												
α	−0.1514***	−5.94	−0.2235***	−7.30	−0.2501***	−7.30	−0.2883***	−18.02	−0.6341***	−31.79	−0.7510***	−25.72
α'	0.2200***	8.98	0.2702***	9.48	0.4174***	11.97	0.4377***	20.88	0.7136***	26.42	0.9297***	53.26
β	0.0184***	6.54	0.0245***	7.25	0.0394***	9.11	0.0405***	18.23	0.0825***	28.60	0.0987***	53.26
β'	−0.0308***	−8.89	−0.0384***	−9.55	−0.0537***	−10.84	−0.0558***	−14.65	−0.0814***	−19.98	−0.0958***	−32.47
SETTLE	−0.0054	−1.58	−0.0045	−1.28	−0.0061	−1.58	0.0260***	6.61	0.0030	0.57	−0.0097*	−1.66
DECEMBER	−0.1474***	−14.82	0.0082	1.13	−0.0423***	−8.40	−0.0893***	−15.09	−0.0969***	−8.52	0.0321**	2.45
HOLIDAY	0.0096**	2.11	0.0079	1.33	0.0035	0.59	0.0016	0.23	0.0156	1.63	−0.0063	−0.73
MONTHEND	0.0028	0.53	0.0744***	10.34	0.0001	0.02	0.0016	0.18	0.0116***	1.82	−0.0193***	−3.51
δ	0.0004	0.23	−0.0073***	−2.92	0.0153***	6.28	0.0070***	5.43	0.0077***	2.39	0.0243***	5.90
a_0	0.0035***	14.86	0.0030***	10.94	0.0039***	19.44	0.0022***	8.62	0.0028***	9.81	0.0016***	5.72
a'_0	−0.0025***	−10.49	−0.0023***	−8.60	−0.0032***	−17.06	−0.0014***	−5.89	−0.0016***	−5.54	−0.0006**	−2.35
a_1	1.0071***	30.86	1.0178***	30.07	1.0206***	23.65	1.0918***	20.19	0.9753***	14.71	1.0359***	22.67
SETTLE	0.0005	1.26	0.0001	0.36	0.0001	0.52	0.0011***	2.85	−0.0011***	−5.65	0.0005***	2.68
DECEMBER	0.0181	1.39	0.0012**	2.59	0.0018***	3.04	−0.0003	−1.37	−0.0006***	−1.95	−0.0002	−0.46
HOLIDAY	−0.0013***	−6.00	−0.0006**	−1.98	−0.0004	−0.56	−0.0006***	−2.56	0.0008	1.42	−0.0005	−1.47
MONTHEND	0.0009	1.03	0.0070***	7.49	0.0001	0.18	0.0040***	4.11	−0.0007***	−2.41	0.0006	1.36
Joint test	619.961***		495.593***		1231.165***		1023.192***		1358.195***		30220.560***	
Panel B: Joint Wald tests using multivariate ARCH-M equations on sample data from February 1984 to July 1997												
Pure expectations (all coefficient = 0)								14,226***				
General expectations (slope and calendar coefficient = 0)								2462.4***				

Note: The model is: $R_{t+n} - Y_t(n) = \alpha_n + \alpha'_n\text{DUMMY} + \beta_n Y_t(n) + \beta'_n Y_t(n)\text{DUMMY} + \phi\text{SETTLE} + \rho\text{DECEMBER} + \omega\text{HOLIDAY} + \mu\text{MONTHEND} + \delta \log h_t + \varepsilon_t$ with R_{t+n} equals to the average overnight rate for n days and $Y_t(n)$ equal to the term repo rate of maturity n at time t. The null hypothesis is $\alpha = 0$ and $\beta = 0$.

The variance is specified as: $h_t = a_0 + a'_0\text{DUMMY} + a_1\sum_{\tau=1,4} w_\tau \varepsilon^2_{t-1} + c\text{SETTLE} + d\text{DECEMBER} + f\text{HOLIDAY} + g\text{MONTHEND}$ with $w_\tau = (6-\tau)/15$ to weight the past innovations across the days of the week as in Engle et al. (1987). DUMMY equals one after December 1991 and zero otherwise; SETTLE equals one on settlement Wednesday and zero otherwise; DECEMBER equals one if the repo matures across the end of the year and zero otherwise; HOLIDAY equals one on the day before a market closed holiday and zero otherwise; MONTHEND is one at month end and zero otherwise. Standard errors are Hansen and Hodrick (1980) robust standard errors with the lagged error equal to the term of the repurchase agreement for each maturity.

*,**,*** Significant at the 10%, 5%, and 1% level, respectively.

Table 7
Weekly spreads between 3- and 6-month T-bill yields

Time period	Average spread (basis points)	Standard deviation
Overall	15.92	16.88
1960s	18.79	13.15
1970s	22.80	19.43
1980s	14.64	17.83
1990s	10.81	13.76
2000s	3.55	13.36

Note: The period from October 1979 to October 1982 is excluded from the spread data because this period was highly volatile due to the Federal Reserve's experiment with targeting M1. The data for the 2000s extends through January 2003.

1990s. This suggests that Longstaff's results, although not representative for data since 1960, could represent a new equilibrium at the short end of the term structure.[11]

To explore the expectations hypothesis during the continued low volatility period following Longstaff's (2000) sample, we re-estimate Eq. (1) on short-term repos rates from November 1999 through January 2003. The data are for overnight, 1-week, 2-week, and 1-month repo rates and are collected on a daily basis from Bloomberg. The results from estimating Eq. (1) on this additional repo data support pure expectations for 1-week and 2-week repos, but reject the expectations hypothesis for 1-month repos.[12] For robustness, we add the calendar-time-based dummy variables as in Eq. (2) and re-estimate our results on the additional repo data. Although we find several significant calendar-time parameter estimates, the results for α and β are unchanged.[13] These results suggest that Longstaff's results may suggest a new market equilibrium, but that his strong support for the expectations hypothesis at all terms up to 3 months is likely sample specific.

[11] Lange et al. (2003) suggest there has been an increase in monetary policy transparency in recent years. Thus, to the extent that monetary policy no longer surprises the market, one would expect money-market rates to be less volatile. We thank an anonymous referee for pointing out this explanation of the changes in the Federal Reserve Board's policy on communicating monetary policy intentions to the markets.

[12] All results on the additional repo data are not provided in the interest of brevity, but are available on request.

[13] While conducting this additional analysis, we also attempted to assess the generality of the results. Cyree et al. (2003) find a US bank settlement effect in overnight US-dollar-based LIBOR. LIBOR quotations are available in a variety of currencies, and we have no reason to suspect that the overnight rates in other currencies suffer the same idiosyncrasies as fed funds rates. We chose to collect LIBOR for Euros, Australian Dollars, British Pounds, and Japanese Yen. Beginning in 2001, the BBA provides daily rates in terms to maturity of overnight, 1-week, 2-week, and monthly from 1-month to 12-months. We collected rates for terms of overnight, 1-week, 2-weeks, and 1-month and use this data for additional estimations of Eq. (1) with Newey and West (1987) test statistics. The results from Eq. (1) on LIBOR for the four currencies are available upon request. The euro and the British pound estimates support the (pure) expectations hypothesis for each term tested (1-week, 2-week, and 1-month). The results from the Australian dollar support (pure) expectations at 1-week and 2-weeks but reject expectations at 1-month. Finally, the results from the yen support the expectations hypothesis at 1-week, but reject the expectations hypothesis at 2-week and 1-month terms. The results are mixed relative to their support for the expectations hypothesis and further suggest that Longstaff's (2000) strong support for this hypothesis does not generalize to other time periods or rates.

5. Conclusions

Longstaff (2000) finds support for the expectations hypothesis at the very short end of the repo term structure. However, the very short end of the repo term structure is known to contain calendar-time-based regularities, and Downing and Oliner (2007) find that these regularities cause the rejection of the expectations hypothesis at the very short end of the commercial paper term structure. Accordingly, we revisit Longstaff's results to determine their robustness.

Using Longstaff's (2000) primary method on in-sample data (but from an alternative source) from May 1991 to July 1997 that covers most of Longstaff's sample period, we reject the expectations hypothesis for one- and 2-week only. When using pre-sample repo rates from February 1984 through May 1991, we reject the expectations hypothesis for each term. When combining the data in a regime-shift specification, we find evidence that our subset of the Longstaff (2000) sample period was less volatile and had higher intercept and lower slope estimates than the pre-sample period. However, the pre-sample results do not provide any insights into whether the pre-sample results or Longstaff's results are more representative of normal market conditions. Additional analysis suggests that the time period covered by the pre-sample repo data is close to average for the period from 1960 to 2003, which, in turn, indicates that the results from the pre-sample data are more representative of historically normal market conditions. Spread volatility on data from a more recent time period following the Longstaff period show volatility is more similar to the Longstaff sample period. This lower volatility in the most recent period hints that Longstaff's results could represent a new interest rate environment, but even with a new interest rate environment Longstaff's strong support for the expectations hypothesis appears to be sample specific.

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